Closest Binary Search Tree Value (medium)

Problem Statement

Given a binary search tree (BST) and a target number, you need to find a node value in the BST that is closest to the given target. A BST is a tree where for every node, the values in the left subtree are smaller than the node, and the values in the right subtree are greater.

Examples:

Example 1:

Input:

```
    Tree:
       5
     /   \
    3     8
   / \   / \
  1   4 6   9
```

Target: 6.4

Expected Output: 6

Justification: The values 6 and 8 are the closest numbers to 6.4 in the tree. Among them, 6 is closer.

Example 2:

Input:

```
    Tree:
       20
     /    \
    10     30
```

Target: 25

Expected Output: 20

Justification: 20 and 30 are the closest numbers to 25. However, 20 is closer than 30.

Example 3:

Input:

```
    Tree:
       2
     /   \
    1     3
```

Target: 2.9

Expected Output: 3

Justification: 3 is the closest value to 2.9 in the tree.

Try it yourself

Try solving this question here:

Java

```java
// Define a TreeNode structure with left and right children.
// class TreeNode {
//     int val;                 // Value stored in the node.
//     TreeNode left;           // Reference to the left child.
//     TreeNode right;          // Reference to the right child.

//     // Constructor to initialize a new node with a value.
//     TreeNode(int x) {
//         val = x;
//     }
// }

public class Solution {

    // This function finds the value in the BST closest to the target.
    public int closestValue(TreeNode root, double target) {
        int closestVal = root.val;
        // ToDo: Write Your Code Here.
        return closestVal;
    }
}
```

Test Case | Run Code Result | Stdout

```
[4,2,6,1,5,7]
3.5
```

Use Example Testcases | Run | Submit

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Solution: Kth Smallest Element In A BST (Medium)

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